Redux Toolkit - Complete Course Notes

Introduction

Redux Toolkit is the official recommended way to write Redux. It helps manage shared application state in a centralized and scalable way.

What is State?

State is data that changes over time. Examples include cart items, logged-in user, theme, language, notifications and todos.

Props Drilling

Passing props through multiple intermediate components just to reach a deeply nested component. It becomes difficult to maintain in large applications.

Context API

Useful for small global data like theme or authentication. As applications grow, business logic becomes difficult to organize in one context.

Why Redux Toolkit?

Redux Toolkit centralizes shared state, organizes logic into slices, improves scalability and debugging, and keeps projects maintainable.

Redux Store

The Store is the single source of truth. Every shared piece of application data is stored here.

Slice

A Slice manages one feature of the application. Example: User Slice, Cart Slice, Todo Slice, Theme Slice.

initialState

Defines the starting structure and default values of a slice.

Dispatch

dispatch() sends an action to the Redux Store requesting a state update.

useSelector

Reads data from the Redux Store inside React components.

Folder Structure

Create app/store.js, features/todo/todoSlice.js, components, App.jsx and main.jsx.

Reducers vs Reducer

reducers contains multiple reducer functions. todoSlice.reducer is the combined reducer. reducer in configureStore registers slice reducers with the Store.

Practical Todo App

Built a Todo application using Store, Slice, Provider, useDispatch, useSelector, Add Todo and Delete Todo.

Complete Flow

User Action → dispatch() → Slice Reducer → Redux Store → useSelector() → Updated UI.

Images Discussed in the Course

Figure 1

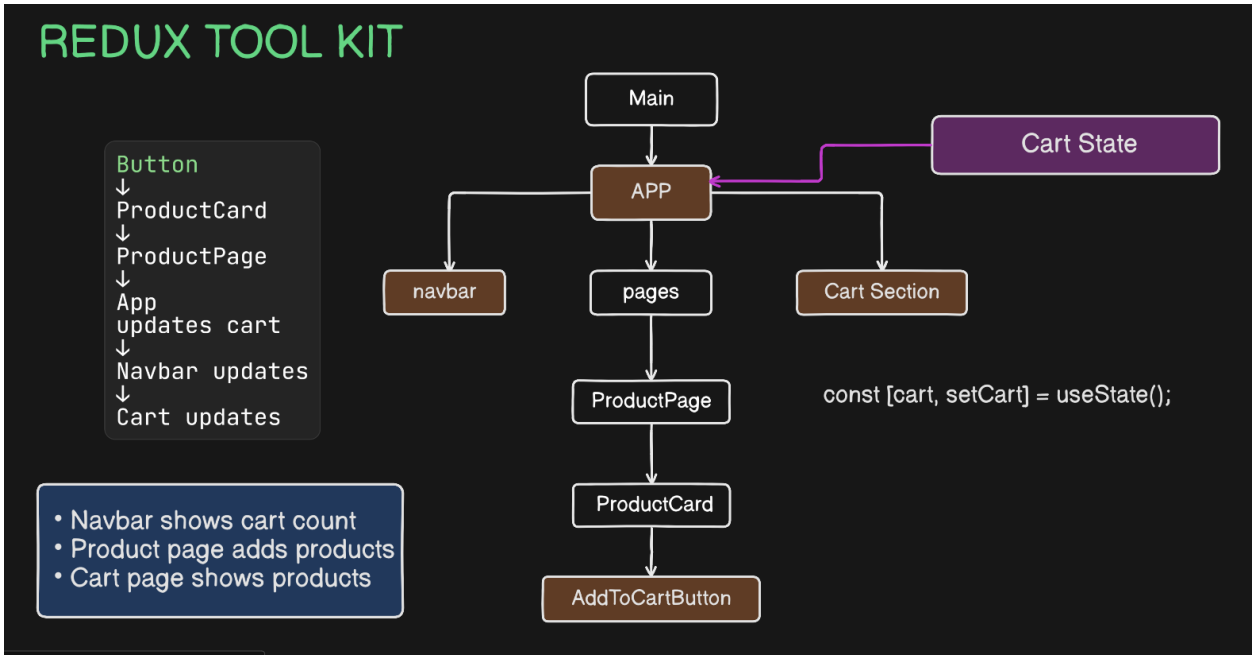


Figure 2

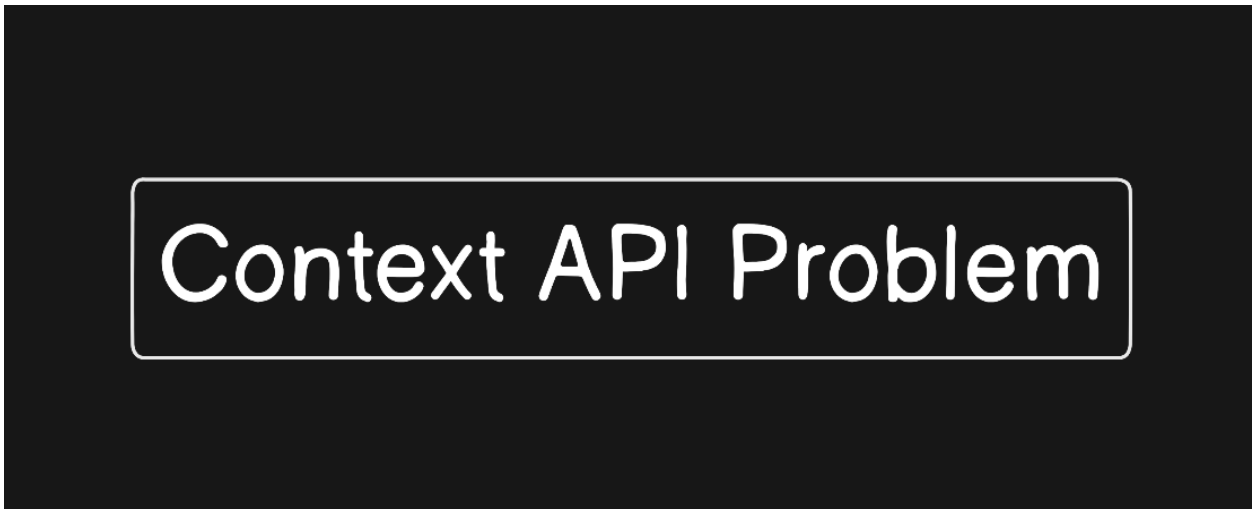


Figure 3

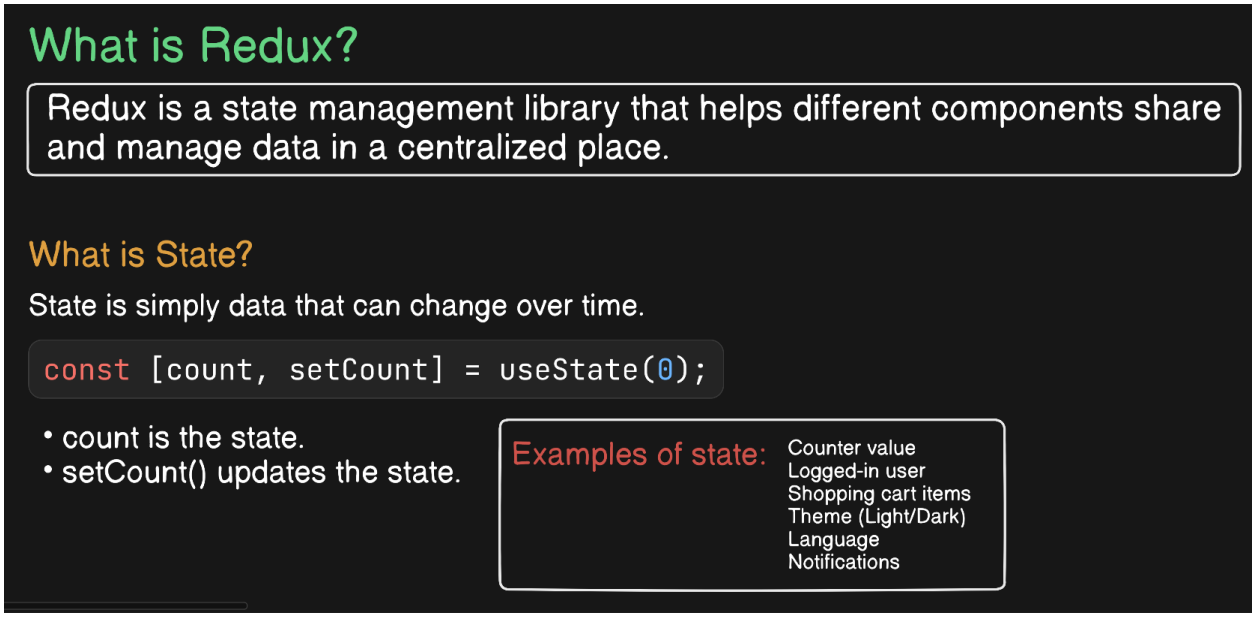


Figure 4

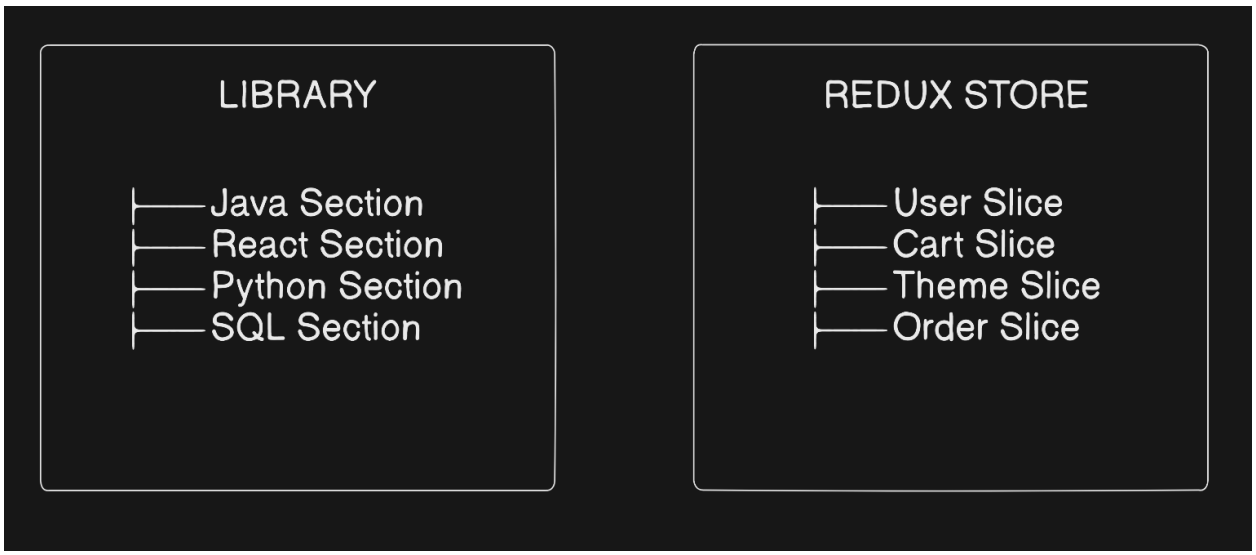


Figure 5

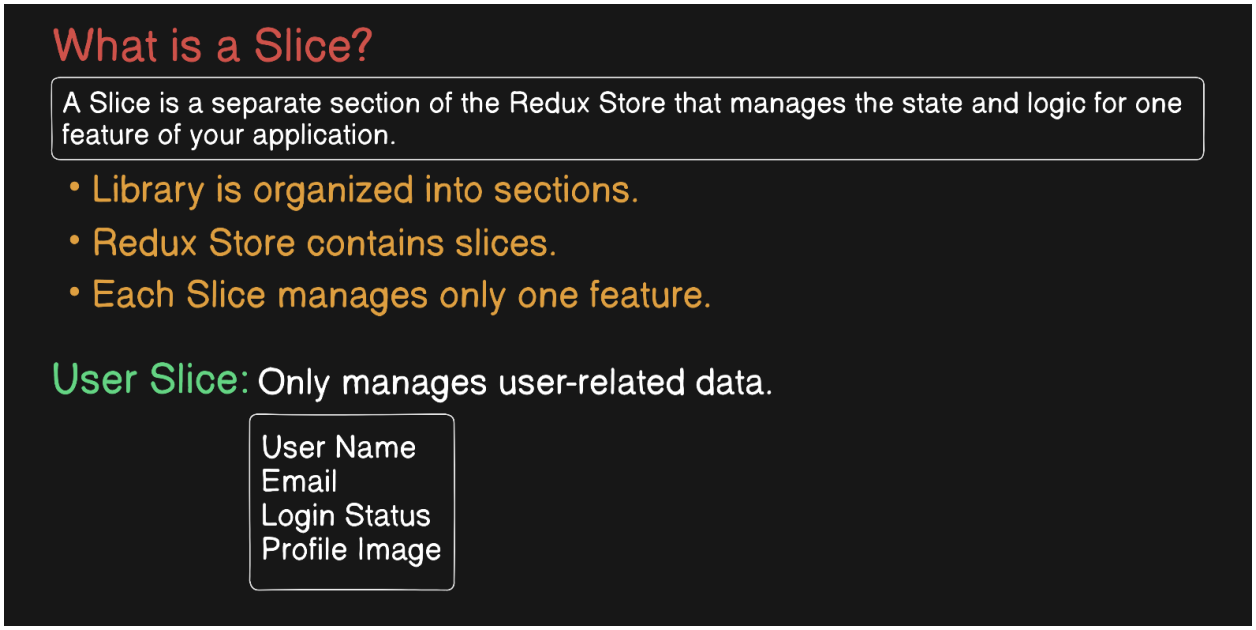


Figure 6



Figure 7

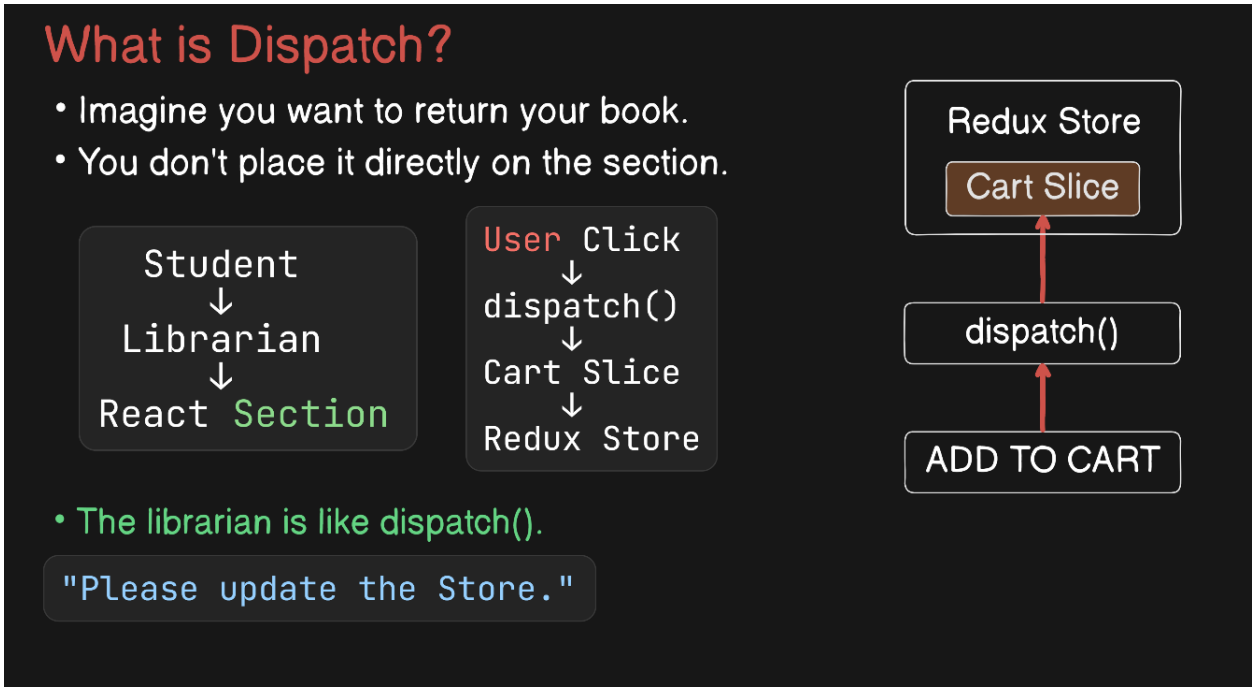


Figure 8

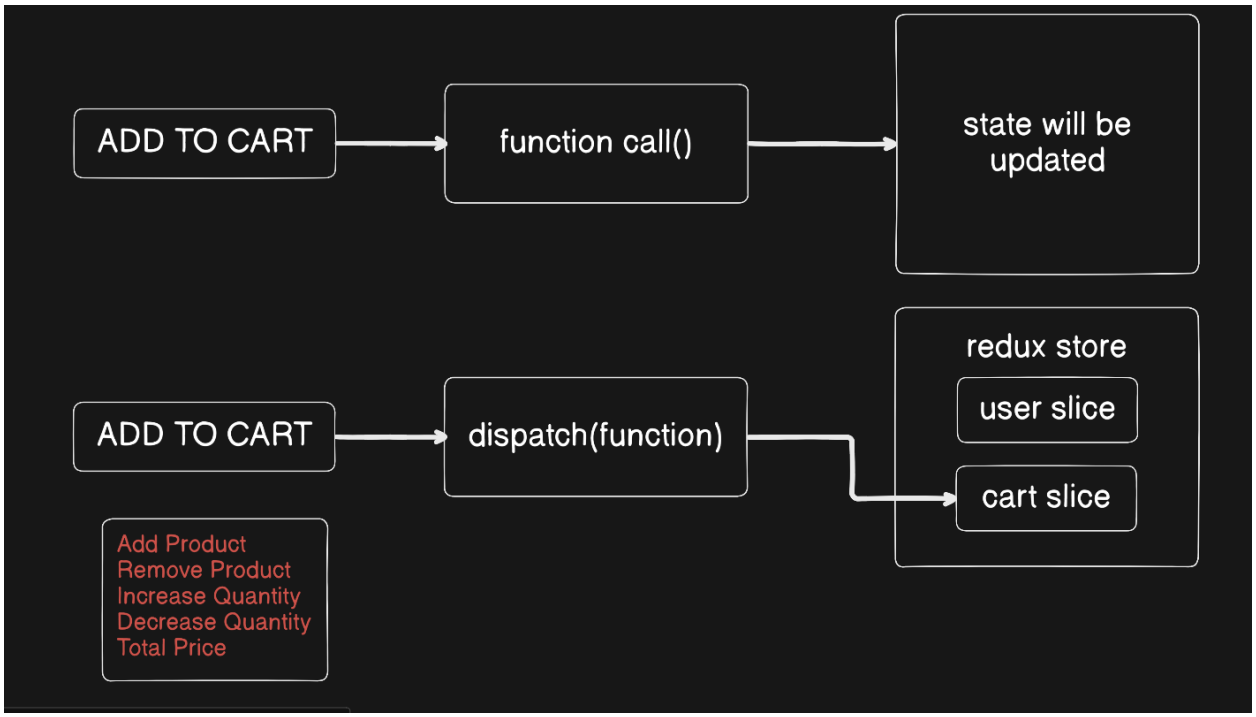


Figure 9

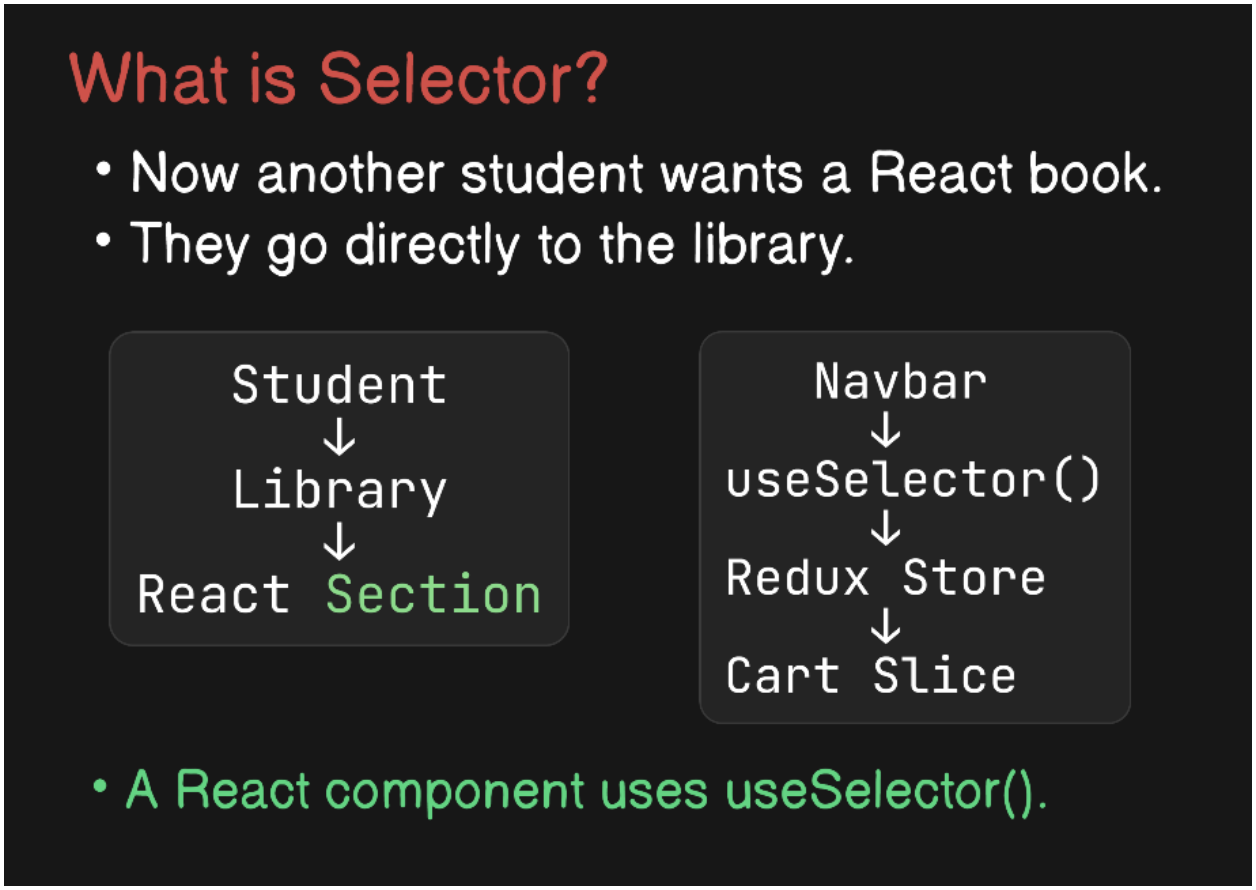


Figure 10

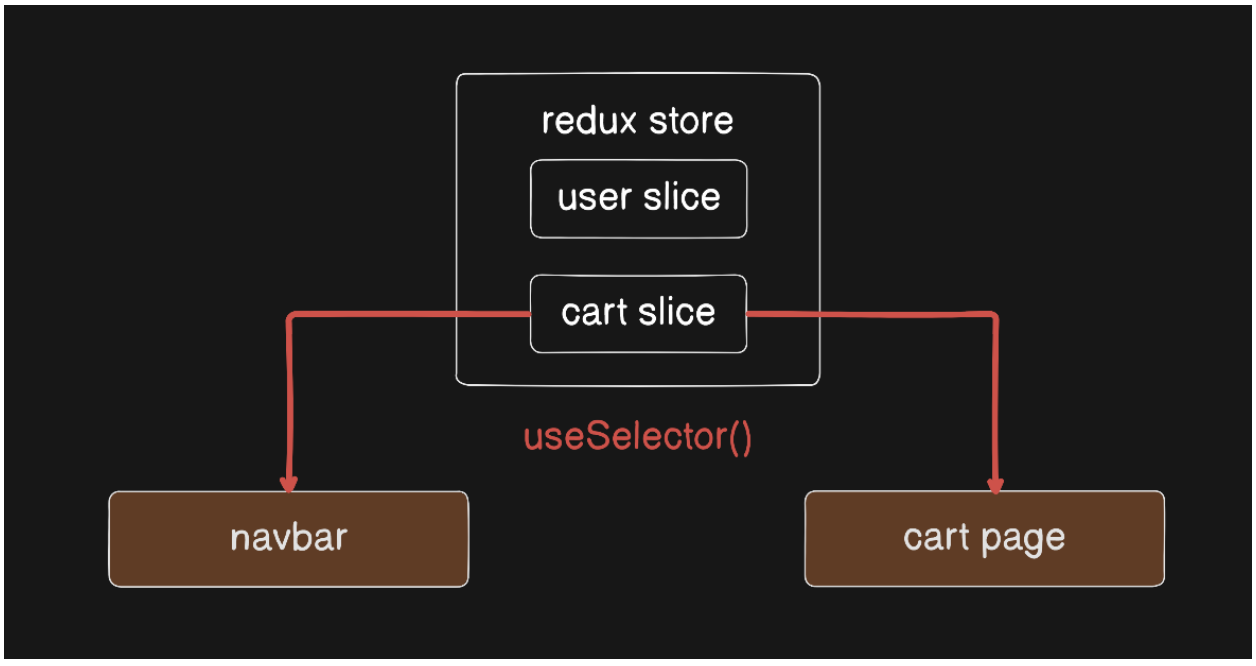


Figure 11

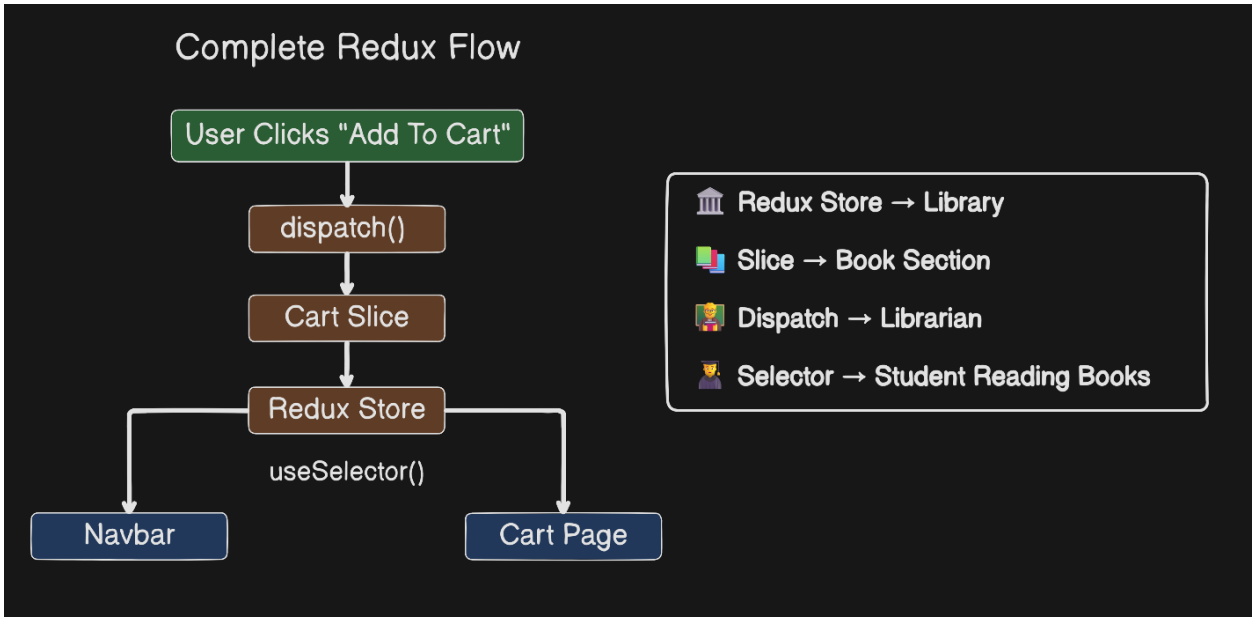


Figure 12

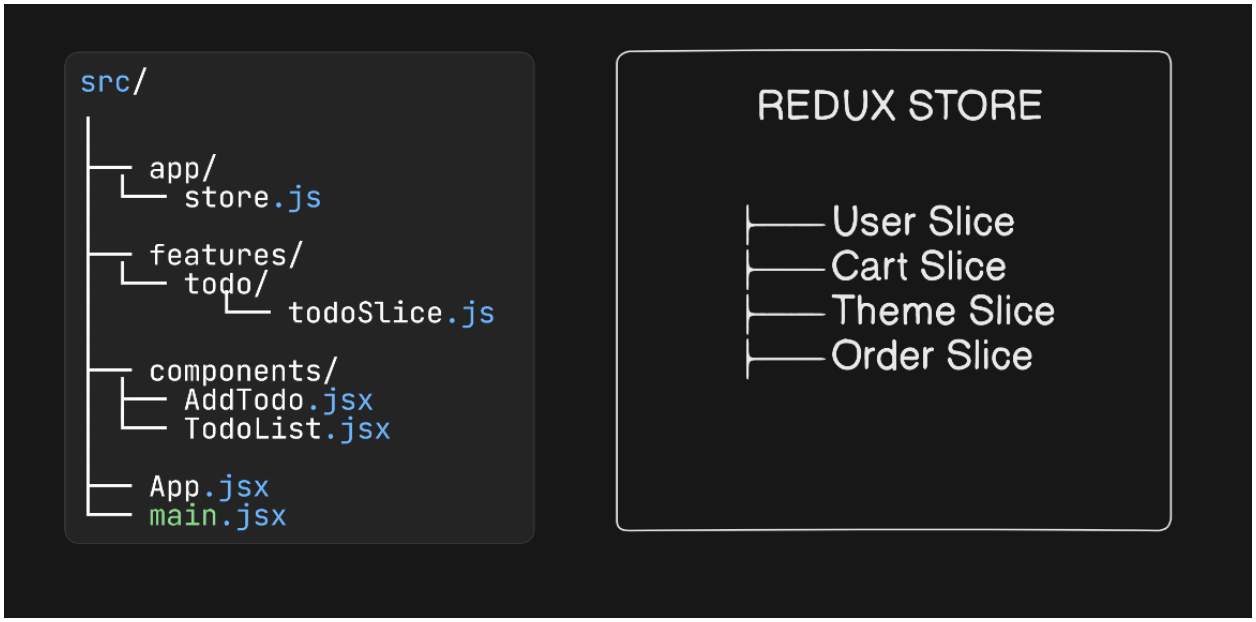


Figure 13



Figure 14

